

ORIGINAL ARTICLE

Applying data science methodologies with artificial intelligence variant reinterpretation to map and estimate genetic disorder prevalence utilizing clinical data

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Abstract

Data science methodologies can be utilized to ascertain and analyze clinical genetic data that is often unstructured and rarely used outside of patient encounters. Genetic variants from all genetic testing resulting to a large pediatric healthcare system for a 5-year period were obtained and reinterpreted utilizing the previously validated Franklin© Artificial Intelligence (AI). Using PowerBI©, the data were further matched to patients in the electronic healthcare record to associate with demographic data to generate a variant data table and mapped by ZIP codes. Three thousand and sixty-five variants were identified and 98% were matched to patients with geographic data. Franklin© changed the interpretation for 24% of variants. One hundred and fifty-six clinically actionable variant reinterpretations were made. A total of 739 Mendelian genetic disorders were identified with disorder prevalence estimation. Mapping of variants demonstrated hot-spots for pathogenic genetic variation such as PEX6-associated Zellweger Spectrum Disorder. Seven patients were identified with Bardet-Biedl syndrome and seven patients with Rett syndrome amenable to newly FDA-approved therapeutics. Utilizing readily available software we developed a database and Exploratory Data Analysis (EDA) methodology enabling us to systematically reinterpret variants, estimate variant prevalence, identify conditions amenable to new treatments, and localize geographies enriched for pathogenic variants.

KEYWORDS

data science, demography, laboratory testing, mapping, population genetics, underserved populations

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1 | INTRODUCTION

Clinical genetic testing has become common in the course of medical practice in the early 21st century with 2.1 million tests billed to Medicare in 2019 representing a 230% increase since 2016 (U.S. Department of Health and Human Services, Office of Inspector General, 2021). Over a five year period over 150 VCH providers ordered a genetic test resulting in over 3000 variants. Like most healthcare systems we previously had no methodology to ascertain, organize and act upon this genetic data outside of the patient encounter. The prevalence of genetic disease is typically derived from national or international multi-disorder databases, disease-specific case series, or registries (Kessel et al., 2021; Nguengang Wakap et al., 2020). However, genetic data are rarely locally available, making it difficult for healthcare and public health systems to gain insight into the regional prevalence of genetic disorders. Geography plays an important role in population genetics as certain disorders are more common amongst groups living in certain geographies such as Zellweger Spectrum Disorder (ZSD) in the Mixteco population of central California (Galarreta et al., 2023). As clinical genetic data is linked to patient demographic data these data may be utilized to gain insights into regional population genetics unavailable elsewhere.

Clinical genetic data often reside in the Electronic Health Record (EHR) in an unstructured format and thus are not readily available. One of the most commonly used methodologies to identify patients with genetic disorders in the EHR is searching based on International Statistical Classification of Diseases and Related Health Problems (ICD) codes. However, ICD coding is inherently error-prone as user-dependent miscoding may confound search results (Marks et al., 2020). Furthermore, many genetic disorders cannot be accurately coded by ICD codes. We solved this problem by obtaining the data for a variant and linked it to a patient's sex, zygosity and demographic data while sorting genes by the inheritance pattern of the disorders associated with them. This allowed us to estimate the genetic disorder prevalence in our catchment area and map such disorders to our local geography utilizing ZIP codes. Herein we report on the methodologies and results of our effort to ascertain clinical genetic variants in our healthcare system, reinterpret the variants with an Artificial Intelligence (AI), estimate genetic disorder prevalence, identify patients with medically actionable genetic disorders and map genetic variants to local geographies.

2 | METHODS

Germline variant data were derived from clinical genetic testing sent from VCH to reference laboratories. Data in the form of discrete variants were derived from the VCH EHRs (Epic and Athena) by either a clinical geneticist or a genetic counselor. Data derived from the EHR was then validated against variant data provided from reference laboratories in the forms of spreadsheets using Microsoft SQL to produce the final variant database. Data were obtained for all available genetic tests sent by VCH providers from January 1, 2016 to January

1, 2021. The data included both proband and familial testing and thus included potentially unaffected family members. The age range of patients in the cohort was estimated by calculating age as of the end of the study (January 1, 2021) as a date of service was not available for all test results. Automated re-interpretation of all variants was attempted utilizing the publically available Franklin© AI (Einhorn et al., 2020, 2023; Einhorn, Kamshov, Einhorn, et al., 2019; Einhorn, Kamshov, Lev, et al., 2019; Einhorn, Kamshov, Paz-Yaacov, et al., 2019; Genoox, 2020; Sheffer, 2002). The Franklin© AI acts as a variant interpretation engine utilizing the ACMG/AMP variant interpretation guidelines (Richards et al., 2015). Franklin© can also utilize phenotypic and familial segregation data to interpret variants, though these functions were not utilized in our study as these data were not available in our database. Franklin's© variant interpretation accuracy was previously validated by comparing its interpretation of 41,267 variants against that of the consensus interpretation of the Canadian Open Genetics Repository (COGR) (Mighton et al., 2022). Comparing Franklin's© and the COGR interpretations based upon a two-tier model (group one benign/likely benign/VUS, group two pathogenic/likely pathogenic) Franklin© and the COGR had a 96.3% concordance. Mighton et al. found that Franklin© had a 94.5% sensitivity and a 96.6% specificity for identifying pathogenic or likely pathogenic (P/LP) variants. Our de-identified variant data was provided to Genoox and variants were run through the Franklin© AI variant interpretation pipeline. If a variant could not be parsed by Franklin©, the original variant interpretation was utilized in the analysis.

Variant data were matched to EHR-derived demographic data by name, age, date of birth (DOB), ZIP code and medical record number (MRN) utilizing structured Microsoft Structured Query Language (SQL). A clinical geneticist analyzed data that could not be matched by SQL and a second attempt was made to match the data manually. ZIP codes were converted to ZIP code tabulation areas (ZCTA) to correspond with the geographic distribution of populations per the 2010 United States Census Bureau (Shukla et al., 2022). Likewise, relevant genetic testing data included laboratory name, test name, copy number variant and/or structural variant (ISCN nomenclature), gene, single-gene variant (HGVS nomenclature), pathogenicity interpretation, and zygosity. Relevant demographic and genomic data were merged to create a database, which was then utilized for EDA in Microsoft PowerBI©.

A choropleth of genomic variation within our data set was generated by mapping variants to ZCTAs generating a choropleth mapping variants to our geographic catchment area in California. Filters for the choropleth were developed within PowerBI© to allow for the selection of variants based upon gene, pathogenicity interpretation, zygosity, ZCTA, and inheritance pattern. Filtering by inheritance pattern was accomplished in PowerBI© by linking the gene data to inheritance patterns per the OMIM inheritance classifications to the EHR-derived variant data (Johns Hopkins University, 2022). Identification of ZCTAs at higher risk for autosomal recessive disorders were identified by mapping homozygous variants regardless of pathogenicity by ZCTA normalized to 2010 US Census data for populations age 18 years and under to best match the VCH pediatric patient population (State of California, Department of Finance, 2010). Due to the

TABLE 1 Estimates of minimum genetic disorder prevalence.

XL Het gene	XL Het prevalence	XL Het total
ABCD1	2.79	13
DMD	2.79	13
MECP2	1.50	7
G6PD	0.64	3
DDX3X	0.43	2
FHL1	0.43	2
BRWD3	0.21	1
CASK	0.21	1
IDS	0.21	1
KDM6A	0.21	1
SV		SV total
arr [GRCh37] 22q11.21 (18919528_21460594) x1		5
arr [GRCh37] 17q12 (34816424_36207539) x1		2
1q21.1q21.2 (145388977_147726541) x1 arr [GRCh37]		2
arr [GRCh37] 21q11.2q22.3 (15484314_48091215) x3		2
arr [GRCh37] 22q13.32q13.33 (48540382_50951420) x1		2
arr [GRCh37] Xp22.33q28 (296520_155228049) x1		2
arr [GrCh37] 10q11.23 (50396854_51365069) x1		1
arr [GRCh37] 10q21.1 (55190013_55573313) x1		1
10q25.1q25.3 (107392689_116166499) x1 arr [GRCh37]		1
arr [GRCh37] 10q26.12q26.3 (123027564_135403394) x1		1
AD Het gene	AD Het prevalence	AD Het total
NF1	10.94	51
COL1A1	7.08	33
FBN1	5.79	27
PTPN11	3.65	17
HBA1	2.79	13
PTEN	2.58	12
TSC1	2.36	11
TSC2	1.93	9
SOS1	1.50	7
HBA2	1.29	6
AR Hom gene	AR Hom prevalence	AR Hom total
PEX6	1.50	7
CFTR	1.29	6
GALC	1.07	5
MCCC2	1.07	5
PAH	0.86	4
SMN1	0.86	4
FANCA	0.64	3
GBA	0.64	3
ATP6V1B1	0.43	2
CCDC40	0.21	1

(Continues)

TABLE 1 (Continued)

XL Hemi gene	XL Hemi prevalence	XL Hemi total
ABCD1	1.72	8
DMD	1.72	8
G6PD	0.64	3
GLA	0.43	2
MID1	0.43	2
PHF8	0.43	2
AVPR2	0.21	1
DLG3	0.21	1
EDA	0.21	1
GJB1	0.21	1
AR Het gene	AR Het prevalence	AR Het total
CFTR	13.30	62
PAH	10.30	48
ACADM	6.01	28
HFE	3.65	17
ARG1	2.36	11
FGFR3	2.15	10
MCCC2	2.15	10
PMS2	2.15	10
HBB	1.93	9
SLC22A5	1.72	8

Note: Counts of the top 10 pathogenic and likely pathogenic (P/LP) variants by gene sorted by zygosity and inheritance pattern. P and LP variants were pooled for the calculation of variant counts. Prevalence is per 100,000 individuals in our 466,011 patient VCH population during our five-year study period. AD het represents individuals expected to be affected with an autosomal dominant disorder, AR Hom affected with a recessive disorder, AR Het autosomal recessive disorder carriers, XL Hemi affected with an X-linked disorder and XL Het carriers for an X-linked disorder. The top 10 P/LP structural variants are also listed. Abbreviations: AD, autosomal dominant; AR, autosomal recessive; Hemi, hemizygous; Het, heterozygous; Hom, homozygous; XL, X-linked.

possibility of incomplete ascertainment of alleles, the count of unique VCH unique patients (466,011) with an encounter for the five-year study period was utilized as the denominator for calculating allele prevalence in Table 1. Additional analysis of allele frequency was performed utilizing 2010 US Census data for ZCTA populations in the select case of the ultra-rare genetic disorder *PEX6*-associated ZSD.

3 | RESULTS

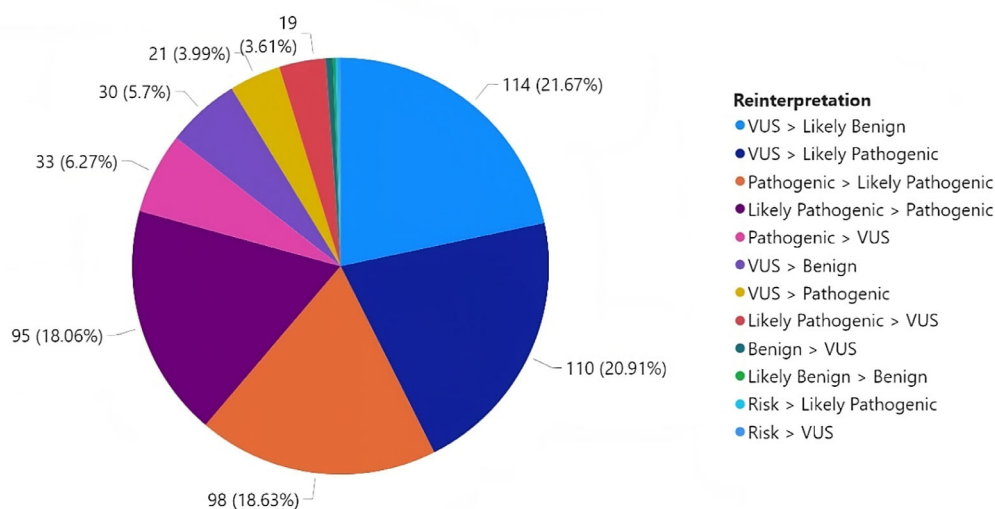
A total of 3044 individuals who had variants were identified and, of these, 98% were linked to geographic data from the EHR. The mean age at time of the end of the study was 10 years with a median of 15 years (range 0 day to 88 years). Testing methodology information was available for 1176 tests which included 613 gene panels, 244 exomes, 170 single-gene, 90 genomes, 33 microarrays, and

18 fluorescent in situ hybridization (FISH) tests. There was a total of 3065 variants of all pathogenicity interpretations: 2474 (80.7%) of variants were monogenic, 584 (19.1%) were structural variants (SV), four (0.1%) were repeat expansion variants and three (0.1%) were methylation variants. Franklin© was also utilized to classify variant type as well as pathogenicity. There were 1704 monogenic variants that could be automatically classified by variant type by Franklin©, including: 52 indels, 762 missense, 26 non-frameshift, 18 synonymous, 10 stop-gain, nine frameshift, one promoter flanking, one start-loss and one stop-loss variant. There were 242 microdeletions and 248 microduplications per Franklin© classification.

Franklin© was able to parse 2688 (88%) of all monogenic and structural variants (SV) with a change in interpretation compared to the

original report for 658 (24%). Franklin© had a high degree of confidence in the interpretation variant pathogenicity for 97% of parsed variants. Two hundred and forty-three (37%) of all reinterpreted variants were upgraded (e.g., benign > pathogenic) and 415 (63%) were downgraded (e.g., pathogenic > benign) (Figure 1). Ninety-one percent (2245) of monogenic variants were parsed and yielded a change in 526 (23%) of variants with 230 (44%) upgraded and 296 (56%) downgraded. Seventy-six percent (443) of SV could be parsed with a classification change in 132 (30%) of variants, and of these variants 12 (9%) were upgraded and 120 (91%) were downgraded. The remaining SVs that could not be parsed were derived from FISH and karyotype results with imprecise nomenclature which is required by the Franklin© AI whereas all chromosomal microarray results were able to be parsed.

Franklin AI Reinterpretation of Monogenic Variants



Franklin AI Variant Reinterpretation of Structural Variants

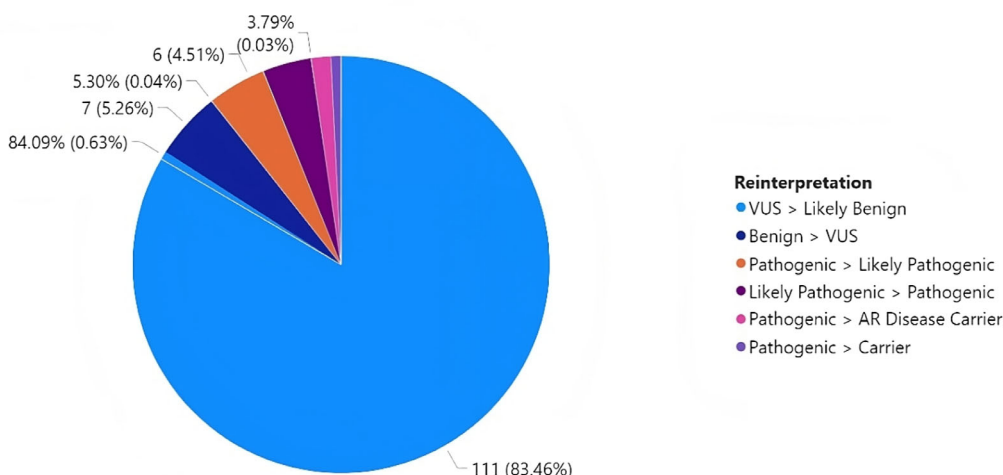


FIGURE 1 Variant reclassifications. Franklin reclassified 658 (24%) of pooled parsable monogenic and structural variants. This figure demonstrates the count of reinterpreted variants with percentages per the Franklin© AI for monogenic (top) versus SV (bottom) variants. There were 156 clinically significant interpretation changes (a reinterpretation to or from P/LP) for monogenic variants while there were no clinically significant changes for SVs. Only variants interpreted by Franklin© as P/LP were included in the final analysis unless there was a discrepancy with the clinician's interpretation as detailed in Results.

Clinically significant Franklin© pathogenicity interpretation changes were defined as a change to or from pathogenic or likely pathogenic (P/LP) to any other interpretation. There were 156 clinically significant interpretation changes in single genes and none for SVs. One hundred and fourteen unique variants were upgraded to P/LP and 42 variants were downgraded from P/LP by Franklin©. There were 22 upgraded variants in genes associated with autosomal dominant disorders considered medically actionable and thus reportable as incidental findings by the ACMG (Miller et al., 2023). A clinical geneticist (J.W.) reviewed all clinically significant changes and there was a 94% (147/156) pathogenicity interpretation concordance rate between the clinician and Franklin©. The interpretation concordance rate for upgraded variants was initially 96% (110/114), though of the five discordant interpretations review of the EHR demonstrated that the patient had a clinical diagnosis of a disorder associated with a pathogenic variant in the gene of interest in four cases resulting in a concordance rate of 99% (113/114). The interpretation concordance rate between Franklin© and J.W. for variants Franklin© downgraded was 88% (37/42) and J.W. upgraded from the Franklin© interpretation as these variants had recently been reported as P/LP and the EHR contained phenotypic information consistent with the variants being pathogenic. As such J.W. and Franklin© for five variants with Franklin© downgrading five variants that J.W. interpreted as P/LP. Data from the Franklin© interpretation was utilized in the final analysis described herein unless there was a discordance with the clinician's interpretation, in which case the clinician's interpretation was utilized for statistical analysis.

A table of P/LP variants filterable by inheritance pattern, pathogenicity, sex, and zygosity was generated in Microsoft Power BI© and demonstrated 739 genetic disorders in our database (Table 1, Figure 2). There were 468 SVs and of these there were 158 P (33.8%), 19 LP (4.1%), 284 VUS (60.7%), 0 likely benign (LB) (0%), and 7 (1.5%) benign (B) variants. The most common pathogenic SV was 22q11.2 microdeletion or DiGeorge Syndrome (arr[GRCh37] 22q11.21

(18919528_21460594)x1) with five affected individuals with the same reported breakpoints. Of the 2467 monogenic variants there were 985 P (40%), 278 LP (11.2%), 1027 VUS (41.6%), 114 LB (4.6%) and 63 B (2.6%) variants. Prevalence in Table 1 was calculated utilizing the number of unique patients with an encounter at VCH during the study period (466,011 patients). As it is likely that not all relevant alleles were ascertained in our student this represents a minimum allele prevalence. There were four repeat expansion variants with the only pathogenic variant being a 52 CAG repeat expansion in *HTT* (Huntington Disease). Of the remaining repeat expansions there were two *FMR1* (Fragile X) premutation alleles and one intermediate allele but there were no full repeat expansions. Disorders of methylation included only Beckwith-Wiedemann Syndrome, with two cases attributed to losses of methylation at imprinting center two and one case of paternal uniparental disomy at 11p15.5. The total five-year minimum prevalence of genetic disorders due to any P/LP variant was 739 per 466,011 (0.16%) unique patients with any encounter at VCH during the study period, and 739 per the 1.4 million patients in the VCH catchment area.

Variants with matching demographic data were mapped to ZCTAs, which allowed for geospatial EDA of clinical genomic data. This generated an interactive map where variants could be sorted by gene, inheritance pattern, interpretation, and zygosity. A choropleth of variant counts normalized to US Census data (population 18 years and younger) was also generated, which allowed for sorting by homozygous individuals to identify geographic areas at higher risk for AR disease (Figure 3). Figure 3 demonstrates areas on the Central Coast and Central Valley with a high prevalence of individuals homozygous for alleles regardless of pathogenicity interpretation. The ZCTA with the highest number of homozygotes normalized for population (variant count/ZCTA population under 18) was 93,274 (Sanger), though none of these variants were LP or P. ZCTA 93257 (Porterville) had the greatest number of homozygotes for P or LP variants (*GBA* x2, *MKKS* x2, *RPE65* x1).

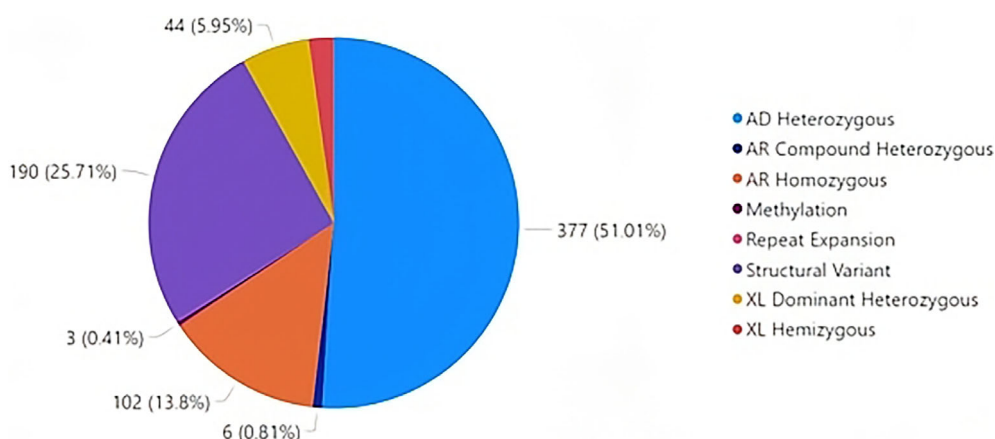


FIGURE 2 Genetic disorder estimation by zygosity and inheritance patterns. Count of P/LP alleles sorted by zygosity and inheritance pattern to estimate the number of individuals affected with an autosomal dominant disorder (377), P/LP structural variant (SV 190), homozygous for an AR disorder (102), compound heterozygous for an AR disorder (6), a female with a dominant X-linked disorder (44), a male with an X-linked disorder (16), methylation disorders (3), and a repeat expansion disorder (1).

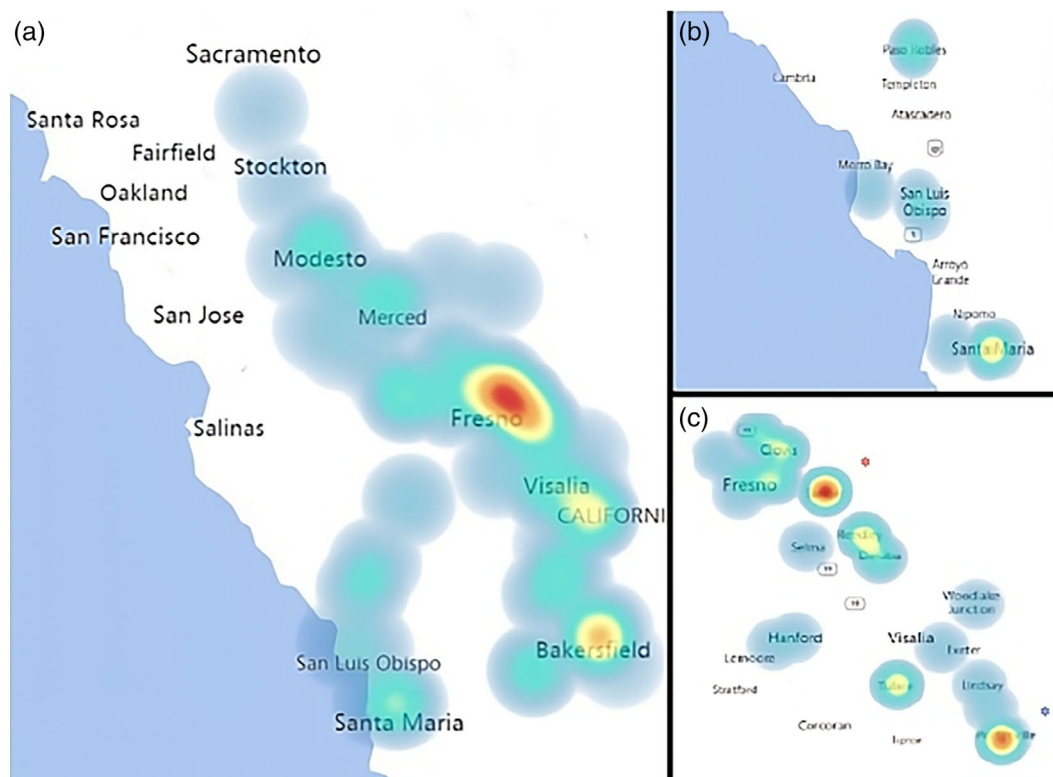


FIGURE 3 Heatmap of homozygous variants. A heatmap of homozygous variants regardless of variant pathogenicity interpretation with total catchment area (a), magnification of homozygous areas in Paso Robles, San Luis Obispo, and Santa Maria where our *PEX6* ZSD patients reside (b), and the small farm communities of Sanger (red asterisks) and Porterville (blue asterisks) with high prevalence of homozygosity.

The most common autosomal recessive (AR) disorder was *PEX6*-associated ZSD while the second most common AR disorder was cystic fibrosis (CF). CF alleles were likely underrepresented in our data set because the California State Newborn Screening (NBS) program routinely genotypes for CF and these results were unavailable to us due to regulatory constraints and clinical management in outside departments. Similarly, the fact that more common AR disorders such as phenylketonuria (*PAH*) and medium-chain acyl-CoA dehydrogenase deficiency (*ACADM*) were less frequent than the ultra-rare *PEX6*-associated ZSD suggests that our methodology did not have complete ascertainment of such alleles. This may be due to the fact that some patients remain undiagnosed, never have genetic testing, or that our data science methodology could not capture such data. Despite these limitations, our methodologies can determine the minimal allele prevalence within a healthcare system, though incomplete ascertainment may exaggerate the differences between the prevalence of different disorders.

Seven homozygotes for a P/LP variant in *PEX6* were identified in which all but one were homozygous for the same pathogenic variant (NM_000287.4:c.1409G>C). EDA mapping of NM_000287.4:c.1409G>C per Figure 4 geographically represented these six affected individuals in Bakersfield, Paso Robles and Santa Maria (ZCTA 93446 x2, 93458 x2, 93306 x1, 93451 x1). Of these individuals the preferred language was Spanish (3), Mixteco Bajo (2) and English (1). The prevalence of NM_000287.4:c.1409G>C per each ZCTA was 18.5, 10.5,

4.9 and 96/100,000 (denominator is US 2010 US Census population 18 years and younger) respectively for a pooled prevalence of 11.8/100,000 for the US Census population per ZCTAs (State of California, Department of Finance, 2010; Steinberg et al., 2006). The minimum prevalence of *PEX6* ZSD due to NM_000287.4:c.1409G>C in the VCH 5-year population data set was 1.3/100,000 (6/466,011) and 0.43/100,000 (6/1.4 million) for our total catchment area. The accuracy of our ZSD patient data was externally validated as each ZSD patient was present as a screen-positive ZSD in our NBS data. Given the high prevalence of ZSD in the Mixteco population we partnered with local health organizations and prenatal genetic laboratories to raise awareness of the diagnosis and include the common Mixteco *PEX6* variant on genetic prenatal testing.

Seven patients homozygous for P/LP variants in genes associated with neurodevelopmental and obesity syndrome Bardet-Biedl Syndrome (BBS) who were amenable to treatment with the newly FDA approved medicine setmelanotide and had been lost to follow-up were also identified (*BBS7* x2 in ZCTA 93727, *MKKS* x2 in 93,257, *BBS4* x1 in 93274, *BBS9* x1 in 95374). The BBS genes with pathogenic variants in our population diverged from the genes most commonly involved in BBS; per one recent meta-analysis, *BBS1* and *BBS10* account for 23% and 15% of BBS cases respectively (Niederlova et al., 2019). There was no observed geographic clustering of BBS in our data set other than counts of two representing siblings. The total prevalence for BBS in the VCH five-year population data set was

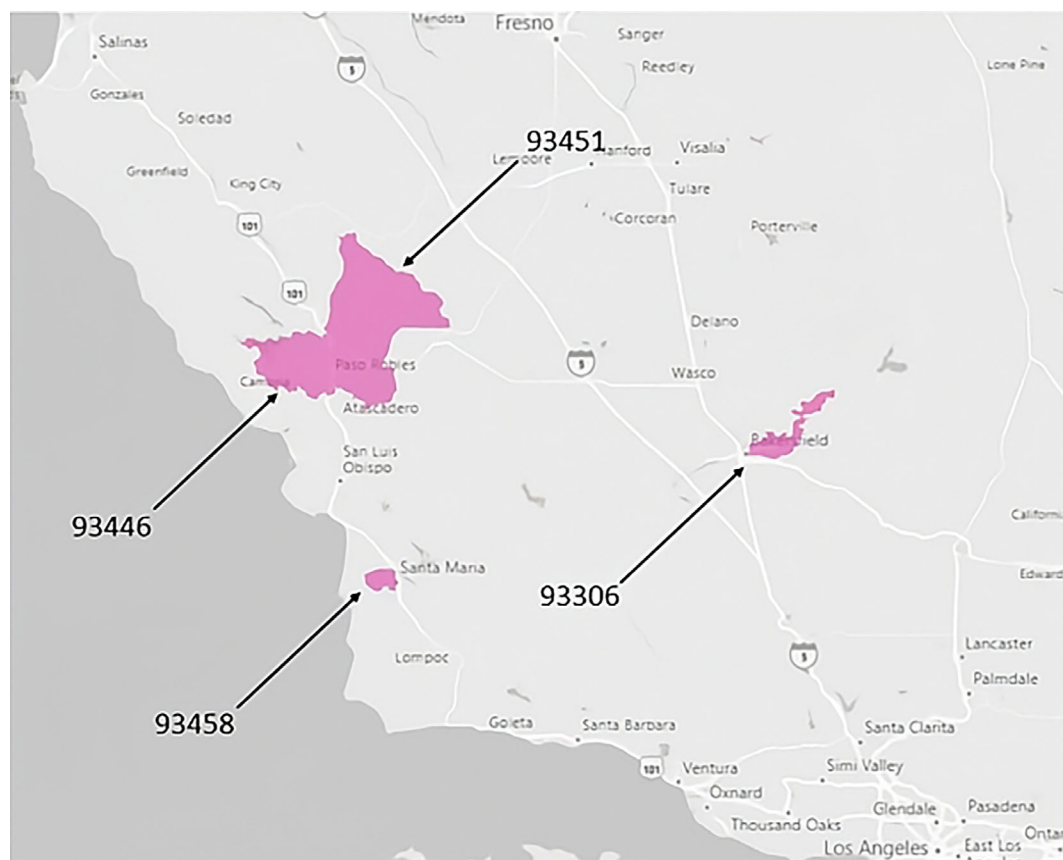


FIGURE 4 Heatmap of homozygotes for NM_000287.4:c.1409G>C. Central Coast ZCTAs with individuals homozygous for NM_000287.4:c.1409G>C (PEX6 pathogenic variant). These are rural ZCTAs with high numbers of indigenous farmworkers per the Indigenous Farmworkers Survey (Mines et al., 2010).

1.5/100,000 (7/466,011) and 0.5/100,000 (7/1.4 million) in the VCH catchment area with a previously reported prevalence of 1/160,000 in northern European populations and up to 1/13,500 in highly consanguineous populations (Gouronc et al., 2020). Setmelanotide treatment was offered to all eligible BBS patients identified, and two of the patients were started on this medication and had significant weight loss with no adverse effects. One BBS patient identified in the database who was lost to all medical follow up for 3 years was identified in the database and was asked to re-establish care with genetics. He was admitted to the hospital the day of his outpatient appointment after previously undiagnosed diabetes mellitus was identified. Additionally, shortly before publication of these findings, trofinetide obtained FDA approval for the treatment of Rett Syndrome (RS). As of the time of publication, two of the RS patients identified by the database were started on trofinetide and two of the BBS patients were started on setmelanotide. While the Franklin® variant classification was used to query for the BBS, RS and ZSD patients, a medical geneticist reviewed all cases and concurred with all of the Franklin® variant classifications. Of the 14 clinically actionable variant interpretation changes in ACMG incidental findings genes two individuals who had a VUS in *VHL* (Von-Hippel-Lindau) upgraded to pathogenic. They were invited to have a genetics consultation and started on a cancer surveillance protocol. The remaining patients with such

clinically actionable variant interpretation changes were offered a genetics appointment in a letter but have yet to follow up as of the time of publication.

4 | CONCLUSIONS

We used a combination of data science and Mendelian genetic principles to facilitate EDA of clinical genetic data utilizing the unique capabilities of AI and choropleths to gain novel insights into our patient population. These methodologies allowed us to estimate the prevalence of genetic disorders within our catchment area, localize geographies enriched for variants, identify patients who were eligible for emerging therapies and systemically reinterpret variants using AI. The fact that we identified multiple patients lost to follow up that were eligible for new therapies or required hospitalization for otherwise unrecognized medical conditions suggests that clinicians and health-care systems have an obligation to manage clinical genetic data in a systemic fashion that makes such data useful well beyond the initial results disclosure visit. Given that patients often change healthcare systems and will also age out of pediatric systems regional or nationwide databases are required to ensure continuity of care for patients with genetic disorders.

Our pipeline was developed with no formal research funding, using only healthcare and informatics professional volunteers along with informatics tools that are often available to many larger healthcare systems such as Epic© and PowerBI©. It is our hope that other healthcare systems could develop similar data pipelines to improve utilization of already available genetic variant data, since our approach is cost-effective and our methodologies did not require programming language skills. At our institution, these data provided actionable and specific insights unavailable elsewhere which are pertinent to the unique patient population served by VCH.

In most institutions, patients are not routinely consented about being re-contacted when variants are reinterpreted and there are typically no healthcare resources allocated for variant database curation and variant re-analysis (Pagan et al., 2021). As such, consideration should be given to counseling patients about the nature of an ongoing, dynamic genomic medicine workup. Geographic mapping of rare variants to small communities may allow for identification of individuals, and could lead to an inappropriate disclosure of Protected Healthcare Information. The Genetic Information Nondiscrimination Act (GINA) protects individual patients with genetic differences against many forms of discrimination, though it is possible that these data could be abused by insurance companies who could decline to provide coverage to geographic areas with a high burden of genetic disorders. As healthcare systems begin to manage and reinterpret variant data, new best practice guidelines will need to be developed to determine how best to steward such data.

Ongoing informatics efforts at VCH include full digital integration of our EHR with reference laboratories utilizing the Epic Genomics Module©. This module will allow for the automatic ascertainment of all genetic testing sent by VCH with near real-time updates of the data into the database described herein. This module also facilitates the development of Best Practice Advisories (BPA) based upon genomic indicators. We are utilizing the prevalence data from this article to prioritize developing BPAs for the genetic disorders most commonly encountered in our population. We are also developing an application program interface (API) to facilitate automatic variant reinterpretation using the Franklin© AI. The clinical variant data management and EDA methodologies we developed can provide potent and actionable insights for healthcare systems that would be rendered even more powerful through collaboration between multiple healthcare organizations, regional newborn screening programs, and national or international organizations.

AUTHOR CONTRIBUTIONS

Suellen Jackson: Data Curation; Formal Analysis; Validation; Visualization; Software; Resources; Methodology. **Rebecca Freeman:** Writing-review and editing. **Adriana Noronha:** Writing-review and editing. **Hafsah Jamil:** Writing-review and editing. **Eric Chavez:** Data Curation; Formal Analysis. **Jason Carmichael:** Writing-review and editing. **Kaylee M. Ruiz:** Writing-review and editing. **Christine Miller:** Writing-review and editing. **Sarah Benke:** Writing-review and editing. **Rosalie Perrot:** Writing-review and editing. **Maryam Hockley:** Writing-review and editing. **Kady Murphy:** Writing-review and editing. **Aimiel**

Casillan: Writing-review and editing. **Lily Radanovich:** Writing-review and editing. **Roger Deforest:** Data curation. **Mark E. Nunes:** Writing-review and editing. **Carolina Galarreta-Aima:** Conceptualization. **Richard Sidlow:** Writing-review and editing. **Yaron Einhorn:** Data curation; Formal Analysis. **Jeremy Woods:** Conceptualization; Data Curation; Formal Analysis; Investigation; Methodology; Project Administration; Resources; Supervision; Validation; Visualization; Writing-original Draft; Writing-review & editing.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to report other than Yaron Einhorn is employed by Genoox, which is the company that owns the Franklin AI.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- Einhorn, Y., Ber, G., Einhorn, M., Lev, O., & Paz-Yaacov, N. (2020). Implementation of ACMG/ClinGen standards for CNV classification using AI: Benchmark results.
- Einhorn, Y., Kamshov, A., Einhorn, M., Paz-Yaacov, N., Lev, O., Yaron, Y., Gross, S., & Lyon, E. (2019). Using artificial intelligence for implementing new recommendations of the PVS1 loss of function criterion.
- Einhorn, E., Kamshov, A., Lev, O., Einhorn, M., Paz-Yaacov, N., Shami, K., ... Gross, S. (2019). Implementation of gene-specific ClinGen variant classification recommendations using artificial intelligence: Frequency thresholds.
- Einhorn, Y., Kamshov, A., Paz-Yaacov, N., Einhorn, M., Bepalov, M., Gross, S., & Lyon, E. (2019). Using artificial intelligence for detection of hotspot regions: ACMG/AMP PM1 criterion.
- Einhorn, Y., Lev, O., Einhorn, M., Trabelsi, A., Paz-Yaacov, N., & Gross, S. J. (2023). Benchmarking an automated variant classification engine (aVCE) algorithm using ClinVar: Results of a time-capsule experiment.
- Galarreta, C. I., Wong, K., Carmichael, J., Woods, J., Tise, C. G., Niehaus, A. D., Schildt, A. J., Verscaj, C. P., & Cusmano-Ozog, K. P. (2023). A homozygous Gly470Ala variant in PEX6 causes severe Zellweger spectrum disorder. *American Journal of Medical Genetics. Part A*, 191, 2057–2063. <https://doi.org/10.1002/ajmg.a.63234>
- Genoox. (2020). Franklin HTML Application.
- Gouronc, A., Zilliox, V., Jacquemont, M. L., Darcel, F., Leuvrey, A. S., Nourisson, E., Antin, M., Alessandri, J.-L., Doray, B., Gueguen, P., Payet, F., Randrianaivo, H., Stoetzel, C., Scheidecker, S., Flodrops, H., Dollfus, H., & Muller, J. (2020). High prevalence of Bardet-Biedl syndrome in La Reunion Island is due to a founder variant in ARL6/BBS3. *Clinical Genetics*, 98(2), 166–171. <https://doi.org/10.1111/cge.13768>
- Johns Hopkins University. (2022). Online Mendelian inheritance in man.
- Kessel, L., Bach-Holm, D., Al-Bakri, M., Roos, L., Lund, A., & Grønskov, K. (2021). Genetic disease is a common cause of bilateral childhood cataract in Denmark. *Ophthalmic Genetics*, 42(6), 650–658. <https://doi.org/10.1080/13816810.2021.1941128>
- Marks, L. R., Nolan, N. S., Jiang, L., Muthulingam, D., Liang, S. Y., & Durkin, M. J. (2020). Use of ICD-10 codes for identification of injection drug use-associated infective endocarditis is nonspecific and

- obscure critical findings on impact of medications for opioid use disorder. *Open Forum Infectious Diseases*, 7(10), ofaa414. <https://doi.org/10.1093/ofid/ofaa414>
- Mighton, C., Smith, A. C., Mayers, J., Tomaszewski, R., Taylor, S., Hume, S., Agatep, R., Spriggs, E., Feilottter, H. E., Semenuk, L., Wong, H., de la Vega, L. L., Marshall, C. R., Axford, M. M., Silver, T., Charames, G. S., Di Gioacchino, V., Watkins, N., Foulkes, W. D., ... Canadian Open Genetics Repository Working, G. (2022). Data sharing to improve concordance in variant interpretation across laboratories: Results from the Canadian Open Genetics Repository. *Journal of Medical Genetics*, 59(6), 571–578. <https://doi.org/10.1136/jmedgenet-2021-107738>
- Miller, D. T., Lee, K., Abul-Husn, N. S., Amendola, L. M., Brothers, K., Chung, W. K., Gollob, M. H., Gordon, A. S., Harrison, S. M., Hersherberger, R. E., Klein, T. E., Richards, C. S., Stewart, D. R., Martin, C. L., & ACMG Secondary Findings Working Group. Electronic address: documents@acmg.net. (2023). ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genetics in Medicine*, 25, 100866. <https://doi.org/10.1016/j.gim.2023.100866>
- Mines, R., Nichols, S., & Runsten, D. (2010). California's indigenous farmworkers.
- Nguengang Wakap, S., Lambert, D. M., Olry, A., Rodwell, C., Gueydan, C., Lanneau, V., Murphy, D., Cam, Y. L., & Rath, A. (2020). Estimating cumulative point prevalence of rare diseases: Analysis of the Orphanet database. *European Journal of Human Genetics*, 28(2), 165–173. <https://doi.org/10.1038/s41431-019-0508-0>
- Niederlova, V., Modrak, M., Tsyklauri, O., Huranova, M., & Stepanek, O. (2019). Meta-analysis of genotype-phenotype associations in Bardet-Biedl syndrome uncovers differences among causative genes. *Human Mutation*, 40(11), 2068–2087. <https://doi.org/10.1002/humu.23862>
- Pagan, J. A., Brown, H. S., Rowe, J., Schneider, J. E., Veenstra, D. L., Gupta, A., Berger, S. M., Chung, W. K., & Appelbaum, P. S. (2021). Genetic variant reinterpretation: Economic and population health management challenges. *Population Health Management*, 24(3), 310–313. <https://doi.org/10.1089/pop.2020.0115>
- Richards, S., Aziz, N., Bale, S., Bick, D., Das, S., Gastier-Foster, J., Grody, W. W., Hegde, M., Lyon, E., Spector, E., Voelkerding, K., Rehm, H. L., & ACMG Laboratory Quality Assurance Committee. (2015). Standards and guidelines for the interpretation of sequence variants: A joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine*, 17(5), 405–424. <https://doi.org/10.1038/gim.2015.30>
- Sheffer, A. (2002). Franklin Variant Interpretation.
- Shukla, K., Seppanen, C., Naess, B., Chang, C., Cooley, D., Maier, A., Divita, F., Pitiranggon, M., Johnson, S., Ito, K., & Arunachalam, S. (2022). ZIP code-level estimation of air quality and health risk due to particulate matter pollution in New York City. *Environmental Science & Technology*, 56(11), 7119–7130. <https://doi.org/10.1021/acs.est.1c07325>
- State of California, Department of Finance. (2010). Census 2010. Retrieved from <https://dof.ca.gov/reports/demographic-reports/census-2010/>
- Steinberg, S. J., Dodt, G., Raymond, G. V., Braverman, N. E., Moser, A. B., & Moser, H. W. (2006). Peroxisome biogenesis disorders. *Biochimica et Biophysica Acta*, 1763(12), 1733–1748. <https://doi.org/10.1016/j.bbamcr.2006.09.010>
- U.S. Department of Health and Human Services, Office of Inspector General. (2021). Trends in genetic tests provided under Medicare part B indicate areas of possible concern.

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